

Trial Day AI Engineer | The Coach That Remembers

This case is for an AI Engineer candidate at OWOW. It tests how you turn a client problem into a useful AI product, make technical decisions, and build a working slice. The scenario and member records below are fictional.

The client brief

Our client runs a leadership programme in which members complete a weekly exercise and commit to a specific action. They receive AI feedback after submitting an exercise. A week later, they may reflect on whether they followed through.

Members have told the client that the feedback can be useful, but each response sometimes feels disconnected from what they committed to previously. The client wants members to feel that the coach understands their progress over time. At the same time, members share sensitive information, so the client needs confidence that feedback uses only appropriate information.

The client asks: **How can the next piece of feedback build on a member's earlier exercise and commitment, while keeping the member's information private?**

We want to see how you clarify this request, decide what should be built first, and demonstrate a small part of your solution. You may question whether every part of the solution needs an LLM.

How the day works

Plan for a maximum of seven hours of case work, followed by a demonstration and technical discussion. Use roughly the first half to understand the problem and shape an approach, and the second half to build a working slice and record your decisions. At the start, you can ask an OWOW colleague questions as you would ask a client or product team. State any assumptions that remain unanswered.

The goal is not to complete a full product. Choose a scope you can explain and demonstrate. We value a sound choice, a working result, and a candid account of what you would do next.

Rules

- Use version control. We would like to see meaningful commits or a small pull request.
- Use a backend stack you can work in effectively. OWOW commonly works with TypeScript and Node.js. Python is also acceptable for the AI service.
- Use React or Next.js for a minimal member-facing demonstration. Visual polish is secondary to a clear, usable flow.
- You may use Mastra, LangGraph, another framework, or straightforward application code. Explain why your choice fits this problem.
- AI coding tools are welcome. Be ready to explain and change the code yourself.

- Use only the fictional data below. Do not commit API keys or other secrets.
- Provide instructions that allow another engineer to run your work. If a model service is unavailable, keep the model behind an adapter, demonstrate the flow with a mock, and explain the real integration.
- You may complete features in any order. Make deliberate trade-offs between breadth and quality.

Fictional data

This fixture gives you three member scenarios and a few records that compete for attention. The records are synthetic and are not SLL member data. Treat each `trustedSessionId` as a server-selected session fixture. The submitted `newExercise` contains no member ID. You do not need to build authentication.

Coaching guidance for this exercise

- Aim for a concise observation grounded in the member's answer, one direct question, and a possible next commitment.
- Use earlier exercise and commitment records only when they help with the current situation. Do not claim that a member completed an action unless the data says so.
- Do not invent personal details, diagnose the member, use therapy language, or fill space with generic praise.
- A response can be useful even when the member has no history.
- Only exercise-visible records belonging to the trusted member may enter this exercise-feedback flow. Private chat and other members' records must be filtered before any model call. A superseded commitment must not be presented as the current commitment.

```

1  {
2    "trustedSessions": [
3      {
4        "id": "session-101",
5        "memberId": "member-101"
6      },
7      {
8        "id": "session-202",
9        "memberId": "member-202"
10     },
11     {
12       "id": "session-303",
13       "memberId": "member-303"
14     }
15   ],
16   "records": [
17     {
18       "id": "ex-101-0",
19       "memberId": "member-101",
20       "type": "exercise",
21       "visibility": "exercise",
22       "week": 0,
23       "text": "I keep taking ownership of every status update instead
of delegating."
24     },
25     {

```

```

26     "id": "com-101-0",
27     "memberId": "member-101",
28     "type": "commitment",
29     "visibility": "exercise",
30     "week": 0,
31     "status": "completed",
32     "text": "I will ask a colleague to run Monday's status
roundup."
33   },
34   {
35     "id": "ex-101-1",
36     "memberId": "member-101",
37     "type": "exercise",
38     "visibility": "exercise",
39     "week": 1,
40     "text": "I postponed a difficult conversation with a colleague
because I wanted more information first."
41   },
42   {
43     "id": "com-101-1",
44     "memberId": "member-101",
45     "type": "commitment",
46     "visibility": "exercise",
47     "week": 1,
48     "status": "active",
49     "text": "By Friday, I will ask my colleague what delivery risk
they see."
50   },
51   {
52     "id": "chat-101-1",
53     "memberId": "member-101",
54     "type": "private_chat",
55     "visibility": "private_chat",
56     "week": 1,
57     "text": "I am thinking about leaving my job."
58   },
59   {
60     "id": "ex-202-0",
61     "memberId": "member-202",
62     "type": "exercise",
63     "visibility": "exercise",
64     "week": 0,
65     "text": "My team could not tell which requests were most
important."
66   },
67   {
68     "id": "com-202-0",
69     "memberId": "member-202",
70     "type": "commitment",
71     "visibility": "exercise",
72     "week": 0,
73     "status": "completed",
74     "text": "I will publish our top three priorities each Monday."
75   },
76   {
77     "id": "ex-202-1",
78     "memberId": "member-202",
79     "type": "exercise",
80     "visibility": "exercise",
81     "week": 1,
82     "text": "I agreed to several urgent requests without asking
what should be delayed."
83   },
84   {
85     "id": "com-202-1",
86     "memberId": "member-202",
87     "type": "commitment",

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```

88     "visibility": "exercise",
89     "week": 1,
90     "status": "superseded",
91     "replacedBy": "com-202-2",
92     "text": "I will decline every unplanned request this week."
93 },
94 {
95     "id": "com-202-2",
96     "memberId": "member-202",
97     "type": "commitment",
98     "visibility": "exercise",
99     "week": 1,
100    "status": "active",
101    "text": "On Monday, I will ask my manager to rank new requests
against our current priorities."
102 },
103 {
104     "id": "chat-202-1",
105     "memberId": "member-202",
106     "type": "private_chat",
107     "visibility": "private_chat",
108     "week": 1,
109     "text": "I had an argument with my manager about my future on
the team."
110 },
111 {
112     "id": "com-404-1",
113     "memberId": "member-404",
114     "type": "commitment",
115     "visibility": "exercise",
116     "week": 1,
117     "status": "active",
118     "text": "By Friday, I will ask my colleague what delivery risk
they see."
119 }
120 ],
121 "scenarios": [
122     {
123         "id": "continuity",
124         "trustedSessionId": "session-101",
125         "newExercise": {
126             "week": 2,
127             "question": "Which conversation or decision did you postpone
this week, and what happened?",
128             "answer": "I postponed the conversation again. I told myself
I needed more data, but I did not ask my colleague what they
thought."
129         }
130     },
131     {
132         "id": "revised-commitment",
133         "trustedSessionId": "session-202",
134         "newExercise": {
135             "week": 2,
136             "question": "Which conversation or decision did you postpone
this week, and what happened?",
137             "answer": "Another urgent request arrived. I paused before
saying yes, but I still have not asked my manager which existing
priority should move."
138         }
139     },
140     {
141         "id": "no-history",
142         "trustedSessionId": "session-303",
143         "newExercise": {
144             "week": 2,

```

```
145     "question": "Which conversation or decision did you postpone  
this week, and what happened?",  
146     "answer": "I avoided telling a teammate that we are likely to  
miss a deadline."  
147   }  
148 }  
149 ]  
150 }  
151
```

What the scenarios are meant to reveal

- **Continuity:** Can the response connect to the relevant earlier commitment without treating an unrelated, completed commitment as the main context or claiming the conversation already happened?
- **Revised commitment:** Does the system recognize the current commitment and avoid treating the superseded wording as the active plan?
- **No history:** Is the response still useful without inventing a past interaction?
- **All scenarios:** Can you demonstrate that private-chat and other-member records never enter the exercise-feedback model input? There is no required wording for the generated feedback.

Features

Read all features before choosing your implementation slice. You can combine them where that makes sense.

As a member, I want feedback that connects to my previous commitment

Select relevant earlier exercise and commitment records for each scenario. Show which records informed the response and why, for example in a development trace or an explanation visible to the reviewer.

As a member, I want useful coaching feedback

Return a concise response with an observation, one question that helps me reflect, and a possible next commitment. Keep the tone direct and practical. Do not invent facts, diagnose the member, or present a generic motivational speech.

As a member, I want my information kept in the right context

Filter records before they enter the model context. Show that another member's record and the private chat record cannot reach this exercise-feedback flow. A prompt instruction alone is not an access control.

As a member, I want a usable experience

Create a small React or Next.js screen where a member can submit the new exercise and view the feedback. Handle loading and failure states clearly. The interface can be simple.

As an engineer, I want to know whether the solution works

Show meaningful checks across the supplied scenarios, including context selection, no-history behavior, and the privacy boundary. Verify what reaches the model, not only what appears in its final answer. Explain how you would evaluate coaching quality more broadly after launch.

If you have time

Choose any extension that helps demonstrate your strengths, such as saving a new commitment, a graceful model fallback, structured traces, a larger evaluation set, or a deployment plan. These are options, not a checklist.

What to hand in

All three deliverables are required. The decision document is a central part of the assessment. Keep it concise: one or two pages, or a well-structured README, is enough.

- 1. **Decision document.** Explain the member problem, the outcome you chose, the questions you asked, the answers you received, and any remaining assumptions. Include a simple architecture or data-flow sketch. Document at least three consequential decisions. For each one, state what you chose, what other approach you considered, why your choice fits this case, what you give up, and what would make you change your mind. Cover your first-release scope, privacy boundary, evaluation approach, failure handling, and next steps.
- 2. **Working prototype and code.** Share a runnable repository with the smallest useful end-to-end slice, setup instructions, meaningful commits, and the tests or checks you used. Mark clearly what is implemented, mocked, and unfinished.
- 3. **Demonstration and technical discussion.** Spend about 10 minutes showing the member flow and then discuss your decisions with the OWOW team for about 20 to 30 minutes. We will ask why you chose your approach and why you did not use other plausible tools or designs. Be ready to explain the trade-offs from your document, respond to a changed requirement, and describe what you would do differently with more time or evidence.

You may format the decision document as you like. A compact table can help:

Decision	Alternative considered	Why this choice fits now	Trade-off or risk	When I would revisit it
Your choice	Another realistic option	Your reasoning	What you give up	New evidence or changed constraint

We assess whether your choices are deliberate and defensible. A polished demo without clear reasoning is incomplete.

How we evaluate the case

We assess problem framing and client communication; justified product and technical decisions; working engineering and code quality; AI output quality and evaluation; and privacy, reliability, and ownership. We assess the quality of the working prototype and the reasoning behind it. The decision document and end-of-day discussion show whether you understand your choices and can defend them when challenged. We will probe alternatives you did not choose to understand when your approach is appropriate. We do not score the number of lines of code or features completed.